Making the Count Exact: Ordinal Vertebral Localization from Overlapping C-arm Exposures, With No Learned Component in the Count

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Data sharing: every imaging source is public. The counting tool and the result files from which Tables 1 and 2 are generated are released with the code.

Abstract

Background. Wrong-level spine surgery persists under the Universal Protocol because the protocol verifies a site that has been *marked*, and a vertebral level has no mark. The level is counted, in theatre, across several limited fluoroscopic fields, and each join between fields can be one body off.

Purpose. To measure how exactly a target vertebra is reached by counting nodes along a chain of overlapping fluoroscopic exposures, linked by image correlation restricted to the step the protocol commanded, with no learned component participating in the count.

Materials and Methods. Retrospective study of public data. Sweeps were cut from labelled long images: 801 abdominopelvic CTs, 76 whole-spine CTs, routine clinical CTs from a second institution (522 reaching the sacrum, 204 holding C2), and held-out real lateral radiographs. Frames of a stated physical span were taken with a commanded step and degraded independently. Consecutive frames were linked by normalized cross-correlation restricted to within 0.6 of a vertebral body pitch of that step; nodes were indexed by ordinal from an anchor, and the target was the k-th node. Nothing is fitted in the count, so no spine was withheld on a model's account. The outcome was the share of walks placing the mark on the body truly k from the anchor, among walks whose target entered a field. **Every headline number uses the annotation's vertebra boxes**; detector-box arms are reported separately.

Results. At three bodies of overlap from C2, 834 of 834 walks were exact on 76 whole-spine studies and 2,640 of 2,650 (99.62 %) on 204 studies from a second institution. From the sacrum, 6,214 of 6,229 (99.76 %) were exact on all 801 abdominopelvic spines, and 7,168 of 7,228 (99.17 %) at two bodies on 522 second-institution spines. Walks used 1.15 to 2.96 exposures. On real radiographs, 95 of 95 walks on 16 spines were exact with the annotation's boxes and 95 of 95 with the deployed film detector's boxes; from C2, 110 of 110 on 24 spines. On 79 cases scored with a render-trained detector's boxes the same walks fell to 187 of 622 (30.06 %), and the node the count started from was the true sacrum on 42.44 % of walks.

Conclusion. Counting across overlapping exposures is exact and needs no network. The residual risk is entirely in the vertebra boxes, and specifically in the anchor.

Abbreviations: NCC = normalized cross-correlation.

Introduction

The Universal Protocol verifies a mark. A vertebral level does not have one. Site marking works because a limb, a side or an organ can be signed before induction and inspected at the time-out; a vertebral body cannot be marked through the skin with the accuracy of a single segment, so the level is established after incision, by counting on intraoperative images that show three or four bodies at a time. Chart review of insured wrong-site cases judged the Universal Protocol would have prevented eight of 13 (62 %) under optimal conditions [1], wrong-site events continued through the protocol era [2], and the one study to assess site-verification protocols in the spine found they failed to prevent about a third of cases [3]. The remedy the spine literature recommends, imaging after exposure against a marked fixed structure [4], is itself another count.

The residue is small per procedure and large per career. Half of responding neurosurgeons in two independent surveys report at least one wrong-level operation [5, 6]. A prospectively collected registry of 34,544 degenerative spine operations recorded 0.12 % (95 % CI 0.09–0.16), with the thoracic spine highest at 0.42 % (CI 0.16–1.08), three times the lumbar and eight times the cervical rate [7]; self-report gives one in 3,110 [5]. Seventeen percent of self-reported events led to legal action or settlement [5], 55 of 68 litigated wrong-disc-level cases ended in a payment [8], the median indemnity for a surgical never event reported to the National Practitioner Data Bank is $75,220 [9], and the mean closed wrong-site claim in a contemporary series was $136,453 with spine surgery the commonest procedure [10]; England's 978 spinal-surgery negligence claims over five years were valued at £535.5 million [11]. The aggregate is bounded rather than counted: 127 reports of wrong surgery reached the Joint Commission in 2024, nine wrong-level and seven of those spinal, through a voluntary system it states captures a small fraction [12], and NHS England recorded 185 wrong-site never events in one year [13]. The often-quoted $1.3 billion is a 21-year cumulative for *all* surgical never events [14], not an annual wrong-site figure, and no defensible worldwide total exists in the evidence assembled here.

What is spent avoiding the error shows what the error is worth. One quaternary centre placed 179 preoperative percutaneous gold fiducials over 38 months, 176 of them thoracic, at a mean of 51.9 minutes and 183 mGy each, purely to mark a level [15]; at a hospital operating-room cost of $34–37 per minute before physician fees [16] that is roughly $1,750–$1,950 of staffed procedural time per patient. Thoracic localization by fluoroscopic counting has been reported at about 15 minutes, falling to about three with a fiducial already in place [17, 18], and that fiducial costs 14.3 minutes of CT-suite time [19].

Two observations organize what follows. First, surgeons do not classify vertebrae by morphology. They count, down from C2 or up from the sacrum, and they disagree even on which structure to count — the pedicle for 67 % and the spinous process for 49 % of respondents in the thoracic spine [20, 21]. The intervention that matches the practice is therefore not a classifier that asserts an

identity, but a mechanism that makes the existing count exact across fields. Second, the automated alternative in the literature imports identity rather than reading it: registering a preoperative CT to the radiograph solves a six-degree-of-freedom pose and transfers the name from a human's reading of the volume [22, 23]. That is unavailable when no CT exists, and a miscounted preoperative scan propagates onto the radiograph faithfully.

This study measures a third thing. Because the C-arm moves a commanded distance between exposures, consecutive limited fields overlap by a known amount and link into one chain of vertebrae; the target is the k-th node from an anchor, no network takes part, and the quantity produced is an ordinal rather than a name.

Materials and Methods

This retrospective study used public, de-identified data only; institutional review determined it not to be human-subjects research.

Why an unconstrained correlation search fails on a spine

The lateral spine is quasi-periodic. Vertebral bodies of similar size repeat at a pitch p of about 34 mm in the lumbar spine, so the normalized cross-correlation between the overlap bands of two consecutive frames, as a function of axial shift Δ, has a peak not only at the true displacement d but also at $d \pm p$, $d \pm 2p$, and so on: a body matched against its neighbour agrees almost as well as a body matched against itself. The margin between the true peak and its neighbour is small, and it is made smaller by exactly the things that differ between two independently taken exposures — gain, offset, collimation, a small rotation, noise. When the alias wins, the two frames are joined one body off, every node above the join shifts by one, and the resulting chain is a perfectly plausible consecutive spine, so nothing downstream detects it. This is the documented dominant failure of automatic vertebral localization; it accounted for all 47 failures in the original three-dimensional-to-two-dimensional registration study [24]. It is also the surgeon's own error, reproduced by a machine.

Why restricting the search to the commanded step removes the competing peak

The protocol does not move the C-arm an unknown distance. It commands a step s — one field span minus two vertebral bodies — and the executed move differs from s by a small residual e, so the true displacement is $d = s + e$. Restricting the NCC search to $|\Delta - s| \le 0.6\,p$ therefore excludes the aliases outright rather than merely outranking them: the competing peaks sit at $d \pm p = s + e \pm p$, and they enter a window of half-width $0.6\,p$ only if $|e| \ge 0.4\,p$, that is, only if the actual move erred by more than about 14 mm in the lumbar spine (Fig. 1). The constraint is prior information injected by the act of moving the C-arm a commanded distance; it is not read off the image, and it does not weaken when the image does. Two consequences are built into the protocol handed to the radiographer. Overlap is set at two bodies, which is enough shared structure to place the peak and few enough frames that there are few links to get wrong. And the last move is a full move or none:

a final step shorter than half a body makes s small, the window around it admits $\Delta = 0$ and $\Delta = p$ together, and the search is back in the situation the restriction exists to prevent. The linker refuses steps under half a body for that reason.

Why an ordinal is a different and safer quantity than a name

Naming a vertebra requires two things: a position in the column, and an alphabet that maps positions to names. The alphabet is a property of the patient. A sixth lumbar vertebra, a thirteenth rib-bearing level or a sacralized L5 shifts every name above the variant by one while leaving the column itself unchanged. An ordinal requires only an anchor and the successor relation. The body four above the sacrum is the body four above the sacrum whether it is called L2 or L3, so transitional anatomy cannot make the ordinal wrong; it can only make a *name* wrong. Nothing is fitted in an ordinal count, so there is no training-distribution question for it, no spine has to be withheld for having been seen, and every scan carrying the anchor was used — 801 abdominopelvic spines, not the 120 a held-out naming evaluation allows.

The algorithm, in the form required to reimplement it

1. Place the anchor at the edge of the first field: the sacral promontory and S1 body at the bottom for lumbar or thoracolumbar work, the odontoid and C2 body at the top for cervical or upper thoracic work. Expose. Do not change field size or magnification for the rest of the sweep.
2. Move the C-arm toward the target by the commanded step s = field span at the spine minus two vertebral bodies, and expose again. Repeat until the target is in the field. Field span at the spine is about 14, 19 and 20 cm for a 9-inch intensifier, a 12-inch intensifier and a 30 cm flat panel, and about 10 cm collimated, given a mobile C-arm's fixed source-to-detector distance [25].
3. Obtain vertebra boxes in each frame from any detector. The count uses their positions only; their classes are not read.
4. For each consecutive pair, compute NCC over the overlap band for $\Delta \in [s - 0.6p,\ s + 0.6p]$ and take the argmax. Refuse the link if the commanded step was under $0.5p$.
5. Map the boxes of frame $i + 1$ into frame i by Δ. A box pairs with the box of frame i whose centre lies within half a pitch of it; unpaired boxes beyond the leading edge become new nodes, appended in order. The ordered union over all pairs is the chain.
6. Index the chain from the anchor node. Two anchor rules were evaluated: *terminal*, in which the anchor is the extreme node at the anchor edge of the first frame, and *class*, in which a classifier identifies the sacrum or C2 and the system refuses when no anchor is found. Every result below uses the terminal rule; the class rule is [PENDING] (arms *sacrum, CTSP, detector boxes, anchor = class* and *sacrum, films, film detector, anchor = class*).
7. The target is node k. Its position in the live frame follows from the accumulated translations.

Data, sweep simulation, and boxes

Serial sweeps of one patient exist in no public collection, so a sweep was cut from each labelled long image: frames of a stated physical width, the step as above, each frame degraded independently by gain, offset, gamma, a small rotation and scale, and a fitted film-noise model, so no two frames share pixels exactly. Sources were 801 abdominopelvic CTs with per-level labels, 76 whole-spine CTs, a second institution's public release of routine clinical CTs (522 of them reach the sacrum and 204 hold C2, and the two sets overlap), and held-out real lateral radiographs (16 reaching the sacrum, 24 holding C2). Overlap was two or three bodies as stated per arm.

Vertebra boxes came from the annotation in every arm except where the table says otherwise. Two detector-box arms were run: a detector trained on renders, scored on the 79 cases held out from it, and the deployed film detector on the 16 real-film spines. Because the count itself is not fitted, an annotation-box arm is not a holdout question; a detector-box arm is, and is restricted to cases held out from that detector.

Outcomes

A walk from the anchor to a target at ordinal k is *exact* when the marked node is the body that is truly k from the anchor in the label sequence. *Reached* walks are those whose target entered a field before the image ran out; walks that did not reach are reported separately and are not counted as errors, because they are a limitation of field rather than of counting. *Anchor held* is the share of walks whose starting node really was the anchor; it is a diagnostic and never a gate. Exposures are the mean number of frames a walk consumed. Compute was measured per frame and per walk. No hypothesis test was applied.

Results

Counting with the annotation's boxes

Table 1 gives every arm. Counting down from C2 on 76 whole-spine studies was exact on 832 of 834 walks at two bodies of overlap and on **834 of 834 at three**, from a mean of 2.24 exposures; the two failures at two bodies fell in one study. On 204 studies from a second institution, 2,646 of 2,654 reached walks were exact at two bodies (99.70 %) and 2,640 of 2,650 at three (99.62 %), the misses confined to two spines in each arm.

Counting up from the sacrum on all 801 abdominopelvic spines was exact on 6,234 of 6,255 reached walks at two bodies (99.66 %) and 6,214 of 6,229 at three (99.76 %), the errors concentrated in six and five spines. On 522 second-institution spines the figure was 7,168 of 7,228 (99.17 %) from 2.96 exposures, with 60 wrong walks in six spines. Walks that never reached their target were 97 and 123 of 6,352 on the abdominopelvic spines and 168 of 7,396 on the second institution's, almost all of them the topmost labelled body or a cervical level far above the sacrum.

Error is not uniform along the column, and it grows with the distance walked. From the sacrum on the abdominopelvic spines, lumbar targets were exact 99.73 % of the time and thoracic targets

95.45 %; on the second institution's spines, lumbar 99.33 %, thoracic 97.41 % and cervical 83.84 %. Counting down from C2 on the same second-institution spines, the gradient runs the other way: cervical 100 %, thoracic 99.71 % and lumbar 97.87 % at two bodies of overlap. Each anchor is most exact near itself.

On real lateral radiographs the count was exact on every walk: 95 of 95 on the 16 spines that reach the sacrum, from 1.61 exposures, and 110 of 110 from C2 on 24 spines, from 1.15. Sixteen and 24 spines are few, and the n is stated beside each figure.

The residual risk is in the boxes, and specifically in the anchor

Table 2 holds the central comparison. On the 79 abdominopelvic cases held out from the render-trained detector, walks with the annotation's boxes were exact on 618 of 618 reached walks (100 %). On the same cases with that detector's boxes, 187 of 622 were exact (30.06 %), and the node the count started from was the true sacrum on 42.44 % of walks. Raising overlap to three bodies did not help — 28.66 % exact, anchor held 39.77 % — because more frames mean more boxes, not better ones. By region, detector-box walks were exact 39.29 % of the time to lumbar targets and 13.60 % to thoracic ones: the further the walk, the more chances a missing or spurious box has to shift the ordinal.

The mechanism of that collapse is the one the method is otherwise built to prevent. A missed box removes a node and renames every body above it by one; a spurious box inserts one and does the same in the other direction. The count is then internally consistent and wrong, which is the surgeon's own failure mode reintroduced by a machine. The anchor is the worst place for it to happen, because an anchor error shifts the entire chain. The fix follows from the diagnostic: identify the anchor by class rather than by taking the terminal node of the first frame, and refuse the walk when no anchor is found. Those arms are [PENDING].

With the deployed film detector on the 16 real-film spines, all 95 walks were exact from 1.47 exposures, with the anchor held on 93.68 %. That the ordinal survived on every walk while the anchor diagnostic did not hold on all of them is not explained here; on those films the terminal node and the annotated landmark need not be the same body, and no arm has yet separated the two effects.

What the count costs, and what naming costs beside it

Linking a walk took 0.10 to 0.29 seconds and decoding 0.0002 seconds, timed in the two large naming arms, which run the same linker. The counting arms load no network, so the encoder's 0.017–0.018 seconds per frame is a cost of naming, not of counting. Against this, repositioning a mobile C-arm takes 36 to 49 seconds per move in the only time-motion measurements available (46 and 49 seconds on test rigs; 41.0 ± 30.9 and 36.4 ± 33.2 seconds to reach a target view) [26, 27], and physical manipulation, not beam time, dominates it. A walk of two to three exposures is therefore one to three minutes of C-arm work and under a second of computation. For comparison, the only clinically translated automatic system reports 4.7 seconds per image on a 2012 graphics card and 17 to 72 seconds per case in its 2018 operating-room deployment [22, 23].

Naming the same chains with the study's encoder and run-level head was correct on 96.25 % of 950 walks on the abdominopelvic spines (lumbar 94.98 %, thoracic 93.75 %) and 94.69 % of 7,405 walks on the second institution's (lumbar 96.21 %, thoracic 91.58 %, cervical 80.35 %). Naming is three to five points below counting on the same sweeps, and the gap widens exactly where the alphabet is least certain.

Discussion

Counting is exact and needs no network. Across 76, 204, 801 and 522 spines the walk from an anchor to a target placed the mark on the right body on 99.17 % to 100 % of reached walks, from two or three exposures, with the annotation's boxes. The mechanism that makes it exact is not a model: it is the restriction of the correlation search to the step the protocol commanded, which removes the periodic column's competing peak from the search space rather than asking a score to outrank it.

The division of labour that follows is this paper's contribution. Counting produces an ordinal, and an ordinal is safe under variant anatomy because it never chooses an alphabet. Naming produces a name, and a name requires an alphabet the patient may not have; that is where the uncertainty lives, and it is a per-spine decision rather than one classification per vertebra. The network's proper job is not to identify each vertebra from its morphology but to decide which alphabet this spine is written in. When the operative plan is stated relative to the anchor — the k-th mobile segment above the sacrum — counting alone suffices and no network is needed at all.

The residual risk sits entirely in the boxes and predominantly in the anchor: the same 79 cases went from 100 % exact to 30.06 % when a render-trained detector supplied the boxes, with the anchor correct on 42.44 % of walks. A method whose accuracy is a function of one box must refuse when that box is absent, and the arm that anchors by class rather than by terminal node is [PENDING]. Until it reports, the deployable claim is narrower than the headline: the linking is solved, the detection is not.

This slots into the Universal Protocol as the site-marking step the spine has never had. Verification enters the target level; the sweep places a digital mark on the body it counts to; the time-out compares that mark's count against the surgeon's own, and a disagreement is resolved by rule rather than by deference. It is assistance, not replacement, and its failure mode is auditable by eye: the chain of overlapping images is a visible object, and a surgeon can see which two frames were joined where. That is a different safety argument from a confidence score.

The deployment argument is architectural, not about speed: the compute is negligible either way, so latency is not the case to make. The case is one model instead of two — no preoperative CT to register against, no optical tracker, no line of sight to maintain, no fiducial placed in a CT suite the day before — running on an embedded computer beside a C-arm the hospital already owns. Community hospitals and ambulatory surgery centres operate without navigation suites, whose capital cost was not measured here and is not claimed; what a registration system needs, a preoperative CT with a correct human count on it, is absent in precisely the setting where the level is hardest to name.

Limitations

The sweeps are simulated. Frames are cut from single long images and degraded independently, which reproduces independent exposure, noise, rotation and scale between frames but not parallax between exposures taken at different C-arm positions and not patient motion during the sweep. Both act on the linking rather than on the count's arithmetic, and neither has been measured. No sweep has been performed on a patient.

Every headline number uses the annotation's vertebra boxes and therefore measures the linker, not a deployable system. The real-film arms are 16 and 24 spines. The second institution's release carries no sixth-lumbar or thirteenth-thoracic class: the ordinal is indifferent to variant anatomy by construction, but that indifference is an argument here, not a measurement on variant spines. Walks that did not reach their target are excluded from the accuracy denominator, and both denominators are in Table 1. One release-level label error sits inside the abdominopelvic errors. Three arms are still running and are marked [PENDING] rather than estimated: anchoring by class on renders and on film, the C2-class anchor detector, and the needle confirmation experiment, in which a metal instrument is re-imaged from the same or a slightly different angle and localized on the map to confirm which body is being touched. Its section in the evidence brief is empty, so nothing about it is reported.

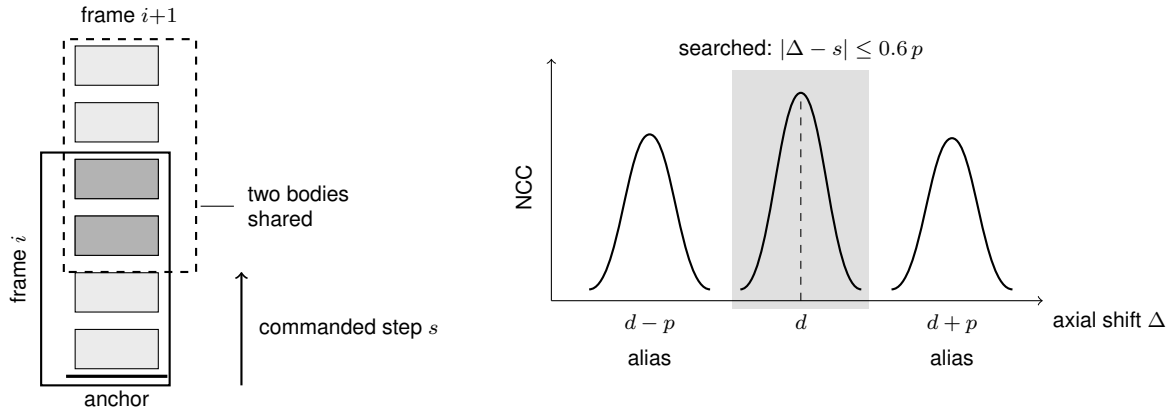


Figure 1: Left: consecutive fields share two bodies after a commanded step s. Right: correlation over axial shift peaks at the true displacement d and again at $d \pm p$, one body pitch away, because the column repeats. Searching only within $0.6\,p$ of s puts the aliases outside the search.

Table 1: Counting arms with the annotation's vertebra boxes. *Reached* walks are those whose target entered a field before the image ran out; the accuracy denominator is *reached*, and both denominators are given so either can be formed. *Wrong* counts walks, with the number of distinct spines they fall in; *Exact* is of reached. Both abdominopelvic arms use all 801 spines. The anchor held on every walk in every arm here.

Arm	Spines	Walks	Reached	Exact	Exposures	Wrong (spines)
C2, two bodies, whole-spine CT	76	834	834	99.76 %	1.99	2 (1)
C2, three bodies, whole-spine CT	76	834	834	**100 %**	2.24	0 (0)
C2, two bodies, second inst.	204	2656	2654	99.70 %	2.05	8 (2)
C2, three bodies, second inst.	204	2656	2650	99.62 %	2.27	10 (2)
Sacrum, two bodies, abdominopelvic	801	6352	6255	99.66 %	1.98	21 (6)
Sacrum, three bodies, abdominopelvic	801	6352	6229	99.76 %	2.42	15 (5)
Sacrum, two bodies, second inst.	522	7396	7228	99.17 %	2.96	60 (6)
Sacrum, two bodies, real radiographs	16	95	95	100 %	1.61	0 (0)
C2, real radiographs	24	110	110	100 %	1.15	0 (0)

Table 2: Where the residual risk is. The first three rows are the same 79 abdominopelvic cases, held out from the render-trained detector, scored with the annotation's boxes and with that detector's. The last two are the same 16 real-film spines. *Anchor held* is the share of walks whose starting node really was the anchor. Every row anchors on the terminal node; anchoring by class is [PENDING].

Arm (overlap)	Boxes	Walks	Reached	Exact of reached	Anchor held
79 CT cases, two bodies	annotation	625	618	100 %	100 %
79 CT cases, two bodies	render detector	625	622	30.06 %	42.44 %
79 CT cases, three bodies	render detector	625	621	28.66 %	39.77 %
16 real films, two bodies	annotation	95	95	100 %	100 %
16 real films, two bodies	film detector	95	95	100 %	93.68 %

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